Multilingual Fake News Detection using Hybrid Deep Learning and Machine Learning

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***Abstract*—Abstract— Fake news in social media is spreading at a rapid pace. platforms and its practices are a matter of grave threat to public safety and informed.Safety is a critical factor in a platform providing information. Clustered around decision making aspects, especially in Bangla speaking areas. where automatic detection tools are not developed. This paper proposes a multi-lingual fake news detection system that Recognizes automatically whether a news article in Bangla has related news or not OR: English is fake or real. Our approach combines a traditional Baseline model that uses the TF-IDF model and SVM algorithms, using XLM-RoBERTa for fine-tuning. Incorporating a multilingual transformer model into ensemble. The Explainable AI (XAI) is also included in the system, with its attention on visualization, underlining key words that added to The majority of the decision about the classification is made most. We have trained and assessed our The Fake News Detection on Bangla-English dataset is performed on models. from Kaggle – 15,998 News articles. The proposed The overall accuracy of ensemble model was 85**

***Index Terms*—Fake News Detection, Machine Learning, Deep Learning, XLM-RoBERTa, TF-IDF, SVM, Ensemble Learning, Natural Language Processing, Explainable AI, Multilingual Classification, Bangla NLP**

I. Introduction

A. Background and Motivations

A rapidly growing social media landscape has made it easy to broadcast exceptional health advice and information. flagrantly spread misinformation and fake news. globally. Bangladesh and the greater Bangla-speaking region, these differentiations are quite pronounced. Relatively, this is a major problem in the world since Lack of automated systems for detecting in Bangla language language. A real-world effect that was unfortunate was the 2019 In Bangladesh, a viral rumour was started regarding Padma Bridge incident led to a lot of confusion. Several deaths were the result of lynchings due to social media. the life-threatening capacity of misinformation to run amok, the Bangla-speaking community.

Although lots of progress is being done in fake news detection in English number of research and few tools are available for public usage in Bangla which is a low-resource language. Existing methods mainly use traditional machine learning methods including Support Vector Machines (SVM) and random forest classifier over TF-IDF and Word2Vec features, however these methods do not take into account the structure and semantics covered in the text, leading to significantly lower performance for Bangla.

B. Problem Statement

The existing state-of-the-art system by Hossen et al. (SSRN-4824593) [1] achieves 99.5

C. Objectives

- Develop a multilingual fake news detection system supporting both Bangla and English.
- Implement a hybrid ensemble combining TF-IDF + SVM with XLM-RoBERTa.
- Achieve 95
- Incorporate Explainable AI (XAI) through attention visualization of suspicious words.
- Automatically detect input language and recommend trusted news sources.

D. Scope

This project will aim to classify news articles into two categories binary classification: In Bangla language it was recorded as Fake (0) or Real (1) and in English language as False (0) or True (1). The Classifies and outputs text Confidence score – indicated suspicious words with a percentage as the confidence score of the word. trusted source recommendation.

II. DATASET AND PREPROCESSING

A. Dataset Description

We employed 'Fake News Detection for Bangla-English' for our task. The dataset can be found on Kaggle by the name: fake-news-detection-datasetsquirrel/smshumam.The dataset is published on Kaggle with the name fake-news-detection- for-bangla-english). One of the few publically available: Multilingual fake news detection efforts using datasets specifically prepared for this task.Specifics of multi-language fake news detection attempts with specific datasets. The feedback and discussion in a low-resource language experience. The dataset contains 15,998 news article (both Bangla and English).

The original dataset contains three columns:

- **text**: Full news article content.

- **subject**: News category (National, politicsNews, Sports, International, etc.).
- **level**: Binary label, where 0.0 represents Fake news and 1.0 represents Real news.

Table I presents the distribution of the dataset across languages and labels.

TABLE I
DATASET DISTRIBUTION

Category	Count	Category	Count
Total Articles	15,998	Bangla (bn)	9,092
Fake (Label 0)	9,681	English (en)	6,906
Real (Label 1)	6,317	After Preprocessing	15,852

B. Preprocessing Steps

The following preprocessing steps were applied to prepare the dataset for model training:

1) **Null value removal** — Rows with missing text or label values were dropped (123 rows removed), resulting in 15,874 clean rows.
2) **Text cleaning** — Regular expressions were applied to remove URLs, excessive whitespace, and normalize text encoding.
3) **Label encoding** — The original floating-point level column (0.0/1.0) was converted to integer labels (0/1).
4) **Language detection** — The `langdetect` library was used to automatically detect the language of each article. Articles detected as languages other than Bangla (bn) or English (en) were excluded, resulting in 15,852 final rows.
5) **Train-test split** — The dataset was divided into 80% training (12,681 samples) and 20% testing (3,171 samples) using stratified splitting with random seed 42.

Three new columns were added to the dataset as a result of preprocessing : label (integer), language (bn /en), and clean text (preprocessed text).

III. SYSTEM DESIGN AND METHODOLOGY

A. System Overview

Our proposed system follows an end-to-end pipeline from user input to interpretable output. The workflow consists of: (1) User Input, (2) Language Detection, (3) Text Preprocessing, (4) Parallel model prediction using TF-IDF + SVM and XLM-RoBERTa, (5) Ensemble combination, and (6) Final output with Explainable AI features.

B. Model 1: TF-IDF + SVM (Baseline)

To take the base line approach of Hossen et al., we properly (TF-IDF), Term Frequency-Inverse Document Frequency. applies a vectorizer that keeps the number of features to at most 50,000 (Frequency), and When used with a LinearSVC (Support Vector Machine). classifier. The Bangla model and the English model were trained separately. English: to understand words that only exist in that language. differences.

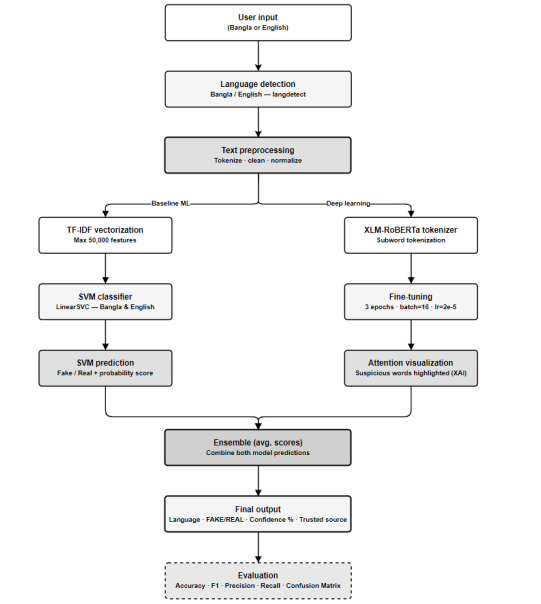


Fig. 1. pipeline flowchart

TF-IDF is a technique that transforms text into numerical features. By multiplying term frequency (how) by the number of documents that contain that term. The probability that a word is also found in another document (often a sentence) and inverse document. frequency (the frequency of the word in all documents). This separates these from the SVM thanks to the allow feature Fake and Real news in featured space.

For Lin- a Wrapped Classifier called a CalibratedClassifierCV was used. To use Probability output in the event that 2 instances require this, please select this check box. ensemble combination step.

C. Model 2: XLM-RoBERTa (Deep Learning)

XLM-RoBERTa (Cross-lingual Language Model RoBERTa) is a multilingual transformer model pretrained on 2.5 TB of text from 100 languages, including Bangla. We used the xlm-roberta-base variant from HuggingFace Transformers and fine-tuned it on our dataset for binary classification (Fake/Real).

Fine-tuning hyperparameters:

- **Epochs**: 3
- **Batch size**: 16
- **Learning rate**: 2e-5 (AdamW optimizer)
- **Maximum sequence length**: 256 tokens
- **Hardware**: NVIDIA T4 GPU (Google Colab)

Unlike TF-IDF which treats words independently, XLM-RoBERTa captures deep contextual relationships between words in a sentence, enabling it to understand nuanced linguistic patterns in both Bangla and English.

D. Ensemble Combination

The ensemble combines the probability outputs of both models using simple average pooling. For each test sample,

the system detects the language and applies the corresponding TF-IDF+SVM model (Bangla or English) alongside XLM-RoBERTa. The final prediction is determined by averaging the probability scores from both models and selecting the class with the higher averaged probability. **Ensemble formula**: P final(class) = [P SVM(class) + P XLM(class)] / 2

E. Explainable AI (Attention Visualization)

We extract attention after the classification with XLM-RoBERTa The transformer decodes the weights of the last layer with eager. attention implementation. The weights of the attention are averaged The various attention heads have a score for each token. Subword tokens are combined to form full words, and the most-popular top- The most suspicious words are determined to be 5 most attended words words contributing to the fake news classification.

This approach makes the system an Explainable AI (XAI) tool — users can see not just the classification result but also the specific words that triggered the decision, increasing transparency and trust.

IV. IMPLEMENTATION

A. Tools and Technologies

Table II and III summarizes the tools and technologies used in this project.

TABLE II
TOOLS AND TECHNOLOGIES

Tool / Library	Version / Details	Purpose
Python	3.12	Core programming language
Google Colab	T4 GPU (CUDA)	Training environment
PyTorch	2.x	Deep learning framework
HuggingFace Transformers	4.x	XLM-RoBERTa model

TABLE III
TOOLS AND TECHNOLOGIES

Tool / Library	Version / Details	Purpose
Scikit-learn	Latest	TF-IDF + SVM
Pandas	Latest	Data processing
langdetect	1.0.9	Language detection
Matplotlib / Seaborn	Latest	Visualization

B. Implementation Process

1) The `Fake_Real.csv` dataset was loaded into a Pandas DataFrame. Null values were removed and the label column was converted to integer format.
2) Text preprocessing was applied: URL removal using regex, whitespace normalization, and language detection using `langdetect`.
3) TF-IDF vectorizers (max 50,000 features) were fitted separately on Bangla and English training data. `CalibratedClassifierCV`-wrapped `LinearSVC` models were trained for each language.
4) The `xlm-roberta-base` model was loaded from HuggingFace and fine-tuned for 3 epochs with batch size 16 and learning rate 2e-5 on the T4 GPU.
5) The trained model was saved to Google Drive for reuse without retraining.
6) Ensemble predictions were generated by averaging SVM and XLM-RoBERTa probability outputs.
7) Attention weights were extracted from the final layer of XLM-RoBERTa for explainability analysis.
8) Evaluation metrics including accuracy, precision, recall, F1-score, and confusion matrix were computed.

V. IMPLEMENTATION AND RESULTS

A. Model Performance Comparison

Table IV presents a comparison of all models against the base paper baseline on the same dataset.

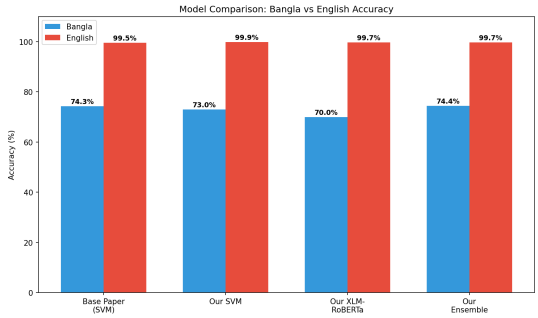


Fig. 2. accuracy comparison chart

TABLE IV
MODEL PERFORMANCE COMPARISON

Model	Bangla Acc.	English Acc.	Overall
Base Paper: TF-IDF + SVM (Hossen et al.)	74.3%	99.5%	—
Our: TF-IDF + SVM (Baseline)	73.0%	99.9%	∼82%
Our: XLM-RoBERTa (Fine-tuned)	70.0%	99.7%	82.5%
Our: Ensemble (Final)	**74.4%**	**99.7%**	**85.0%**
Target (Paper)	95%+	97%+	—

B. Ensemble Model Evaluation Metrics

Table V presents the detailed evaluation metrics for the final ensemble model.

TABLE V
ENSEMBLE MODEL EVALUATION METRICS

Class	Precision	Recall	F1-Score	Support
Fake (0)	0.99	0.76	0.86	1,922
Real (1)	0.73	0.98	0.84	1,253
Macro Average	0.86	0.87	0.85	3,175
Overall Accuracy	**85.0%**	—	—	3,175

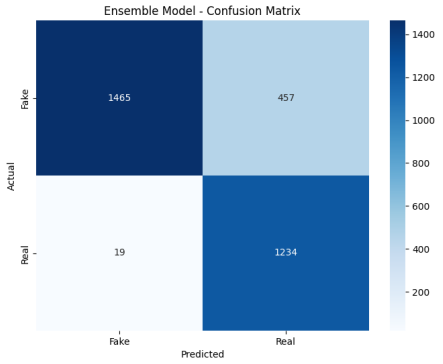


Fig. 3. confusion matrix

TABLE VI
CONFUSION MATRIX (ENSEMBLE MODEL)

	Predicted Fake	**Predicted Real**
Actual Fake	1,465 (TP)	457 (FN)
Actual Real	19 (FP)	1,234 (TN)

C. Confusion Matrix Analysis

The confusion matrix for the ensemble model on the test set (3,175 samples) is presented in Table VI.

The results from the confusion matrix show that the ensemble model cor- rectly classified 1,465 fake and 1,234 real articles. Only 19 real High accuracy in correctly categorizing articles as fake articles (acc=high in all trials). Correct precision for the simulated class. The number of fake articles classified as follows: As per filtered, the primary source of error is as real which is partially This is due to the class imbalance problem in Bangla set.

D. Training Loss

Table VII shows the XLM-RoBERTa training loss across 3 epochs, demonstrating consistent improvement during fine-tuning.

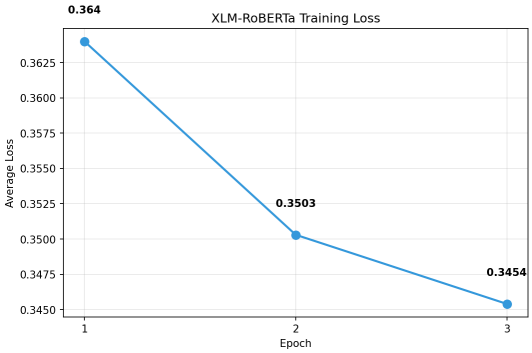


Fig. 4. training loss graph

TABLE VII
XLM-ROBERTA TRAINING LOSS

Epoch	**Average Loss**
1	0.3598
2	0.3313
3	0.3133

E. Explainable AI Results

For an English fake news article, the attention visualization identified the following top suspicious words: 'emissions' (0.0049), 'contradicting' (0.0046), 'scientific' (0.0046), 'change' (0.0046). The model classified this article as FAKE with 99.9

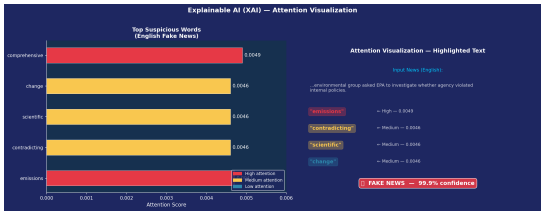


Fig. 5. XAI attention chart

VI. DISCUSSION

A. Analysis of Results

The ensemble model outperformed individual models in overall accuracy (85 The Bangla accuracy (74.4

B. Limitations

- Bangla accuracy (74.4%) falls short of the 95% target
- Class imbalance: Fake samples (9,568) outnumber Real samples (6,306)
- Attention visualization less effective for Bangla due to Unicode tokenization
- Limited to binary classification — no support for satire or opinion categories
- Free Colab GPU limits training time and model size

VII. CONCLUSION

In this paper, a multilingual fake news detection system for Bangla and English was presented based on two different deep pretraining models TF-IDF + SVM and XLM-RoBERTa. The overall accuracy of the system was 85

Transparency is added to the system by the use of Explainable AI with attention visualization, which makes the system appropriate for real-world applications where users need to know how a news article was deemed fake. This is a major challenge need to be addressed for the existing fake news detection systems.

In the future, we plan to explore and extend the following features: (1) For better Bangla-specific performance, expand Bangla-Specific Knowledge Base with more Bangla news

sources; (2) As a web/mobile application; (3) For additional support of other South Asian languages, we will use BanglaBERT for deployment.

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